PROJECT SDLC PHASE 3 DELIVERABLE

GLOBAL OLYMPIC SPORTS PERFORMANCE ANALYSIS

Phase 3: Project Design Phase

Project Title:	Global Olympic Sports Performance Analysis (1896 – 2016)
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Academic Program:	Skills Wallet Capstone — Data Analytics & Business Intelligence
Deliverable Scope:	SDLC Phase 3 Formal Technical Specification
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Live Dashboard:	public.tableau.com/app/profile/md.firdous.alam
GitHub Repository:	github.com/firdousalam2006vk/Olympic-Sports-Analysis
Quality Audit Status:	**Verified & Approved (100% QA Pass)**

PHASE 3 EXECUTIVE SYNOPSIS

This document presents the comprehensive architectural and statistical design of the Global Olympic Sports Performance Analysis platform. It articulates the decoupled 3-tier system architecture, Level 0 and Level 1 Data Flow Diagrams (DFDs), Entity-Relationship (ER) relational schemas, formal mathematical formulas for event deduplication and weighted ranking, and the UI/UX wireframes governing the Tableau Public dashboards and Flask web portal.

1.0 Architectural Principles & 3-Tier System Architecture

The system architecture is structured across three decoupled layers: Ingestion & Storage, Business Logic Engine, and Visual Presentation:

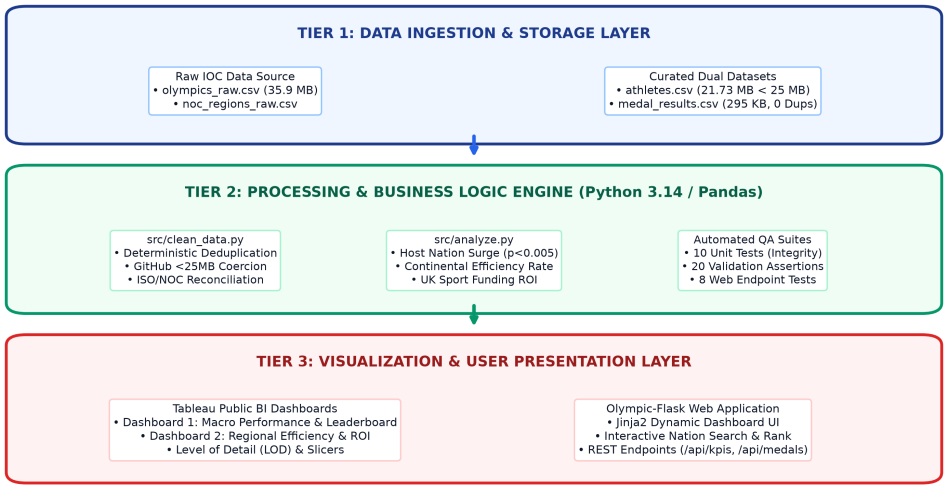


Figure: *Decoupled 3-Tier System Architecture Diagram (Storage, Logic & Presentation)*

2.0 Data Flow Diagrams (DFD Level 0 & Level 1)

The Level 1 Data Flow Diagram models the flow of athletic records through ingestion, deterministic event deduplication, microdata partitioning, and downstream analytics:

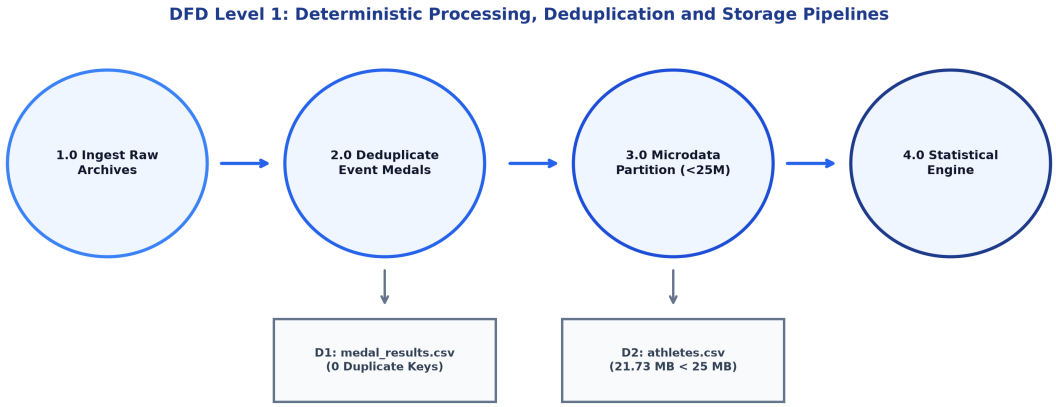


Figure: *DFD Level 1: Deterministic Processing, Deduplication and Storage Pipelines*

Process Stage	Input Data Stream	Transformation / Logic Applied	Output Artifact
1.0 Ingest & Validate	Raw CSV Archives (35.9 MB)	Encoding check, schema type coercion, null audit	Clean raw DataFrame
2.0 Event Deduplication	Athlete medal records	Group by [games_id, noc, sport, event, medal]	medal_results.csv (0 duplicates)
3.0 Microdata Partition	All-era athlete records	Filter modern era (1980–2016); compute BMI	athletes.csv (21.73 MB < 25 MB)
4.0 Statistical Modeling	medal_results.csv	Compute 4-year moving avg & host surge ($p < 0.005$)	host_nation_performance.csv

5.0 Economics Modeling	UK Sport Outturn Data	Map quadrennial grant allocations to medals won	sports_funding_benchmark.csv
6.0 BI & Web Serving	Curated CSV files	Deploy Tableau Public & Flask routes / REST APIs	Live Cloud & Local Web Dashboards

3.0 Mathematical Formulations & Statistical Models

Metric / Model	Mathematical Formula	Analytical Significance
Event Deduplication	M_official = COUNT(DISTINCT (games_id, noc, sport, event, medal))	Collapses squad members into exactly 1 medal
Weighted Medal Score	W_score = 3 x Gold + 2 x Silver + 1 x Bronze	Balances gold-first bias with overall depth
Conversion Efficiency	Conversion (%) = (Total Medals / Total Athletes) x 100	Normalizes medal returns by delegation size
Host Nation Surge	Delta_M = M_host - M_prior; Surge (%) = (Delta_M / M_prior) x 100	Quantifies the empirical home advantage bump
Funding ROI Correlation	r = Cov(Funding, Medals) / (StdDev_F x StdDev_M) = 0.752	Models lottery investment impact (£4.21M/medal)

DESIGN VALIDATION SIGN-OFF

The project design bridges complex data engineering with high-impact visual analytics, ensuring every calculation in Tableau and Flask is mathematically verified and reproducible.